

Tariff or GST?

The fiscal case for rationalising the duty on primary aluminium

A standalone note, for eventual inclusion in Rationalizing Tariff Structures in the Indian Downstream Aluminium Industry. Every figure is reproduced by `tariff_gst_arithmetic.py`, which declares its baseline in one block and reports results across the plausible range of each parameter.

1 The argument, and what it rests on

The objection that stops tariff rationalisation is fiscal: cutting the duty on primary aluminium forgoes revenue. The answer has three parts, and they are of descending robustness — which is how they should be presented.

First, and independent of any data: the two instruments are not substitutes. A customs duty on an intermediate input is a tax on production; GST is a tax on consumption, creditable along the chain. Optimal-tax theory since Diamond and Mirrlees holds that revenue should be raised on final uses and production left undistorted. That is the distinction between these two instruments exactly.

Second, and resting only on an identity: the duty is a very inefficient way to raise the money. Because domestic metal is priced at import parity, the duty lifts the price of *all* metal sold domestically while the exchequer collects only on the imported fraction. The ratio of what is transferred to what is collected is

$$\frac{a}{c} = \frac{\phi(1 - \mu)}{\mu(1 - f)},$$

where μ is import penetration, ϕ the pass-through of the duty into the domestic price and f the share of imports entering duty-free under trade agreements. **No price, exchange rate or elasticity appears in it.** At the baseline the ratio is **17.5 to 1**; across every corner of the parameter ranges declared in §2 it runs from **5.5 to 39**. It is never near unity, and that is the only thing the argument requires.

Third, and resting on more assumptions: the sum at stake is trivial. Net of duty-free imports the duty raises about **INR 444 crore** — 0.02 per cent of FY26 gross GST collections, or under two hours of GST receipts. Replacing it exactly would take **0.18 of a percentage point** on the GST rate applying to these goods.

The mechanism. The duty raises the marginal cost of every downstream fabricator, because metal is bought at import-parity prices whether it is imported or not. Removing it lowers that marginal cost. Since bought-in inputs are 83.1 per cent of downstream output value, the relief is amplified roughly fivefold by the time it reaches value added. That released margin is the space in which a marginally higher consumption tax can be absorbed — and the exchequer then taxes the larger base that results.

2 Baseline

Every number in this note comes from the parameters below. Each is given with a plausible range, and §6 reports how far the conclusions move across those ranges.

	Parameter	Baseline	Range	Source and remarks
B1	Basic customs duty, HS 7601	7.5%	—	Tariff schedule
B2	Social welfare surcharge	10% of BCD	—	Levied on BCD; not creditable
	<i>Effective pre-IGST duty t</i>	8.25%	—	Derived
B3	GST / IGST on aluminium	18%	—	Creditable; does not enter cost
B4	Domestic primary consumption Q_C	4.50 Mt	3.5–5.0	Industry data, CY2024
B5	Primary imports Q_M (HS 7601)	0.392 Mt	—	Report §3.1, 2024
	<i>Import penetration μ</i>	8.7%	—	Consistent with the reported “under 10 per cent”
B6	Value of those imports	US\$1.03bn	—	Report §3.1
B7	Exchange rate	INR 87/USD	83–90	Assumption
	<i>Implied landed price</i>	INR 2.29 lakh/t	—	Derived from B5–B7
B8	Duty pass-through ϕ	1.00	0.70–1.00	Import-parity pricing; $\phi = 1$ is full parity
B9	Duty-free share of imports f	40%	0–70%	The least certain parameter. ASEAN, Japan and Korea rates are zero on most HS 7601 lines
B10	Supply elasticity ε_s	0.20	0.10–0.50	Metals literature; primary supply is capital-constrained
B11	Demand elasticity $ \varepsilon_d $	0.50	0.20–0.80	Metals literature
B12	Metal share of downstream input bill θ	60%	30–70%	Not observable in the ASI
	Amplification $m/(1-m)$	4.91	—	Report §4.7, $m = 0.8309$
	Downstream output (ASI)	INR 2.46 lakh cr	—	Three fabrication classes, 2022–23
	Downstream GVA (ASI)	INR 41,653 cr	—	2022–23
	FY26 gross GST collections	INR 22.27 lakh cr	—	Official monthly releases

Two of these deserve comment. **B9** is the weakest link: the exact share of HS 7601 arriving duty-free is not published in the working data, and the baseline of 40 per cent is an estimate. It is treated as a range throughout, and the central result is reported at $f = 0$ as well. **B12** is not observable in the ASI at all, which is why the induced-expansion calculation in §5.2 is presented as a bound rather than an estimate.

3 Why a duty and a consumption tax are not equivalent

3.1 Production efficiency: do not tax intermediate goods

The governing result is the Diamond–Mirrlees production efficiency theorem.¹ Under standard conditions the optimal tax system preserves productive efficiency: revenue should be raised on final uses, and *intermediate goods should not be taxed*. A tax on an intermediate input distorts the technique of production as well as the level of consumption, so it does two kinds of damage where a consumption tax does one.

¹Diamond & Mirrlees (1971), *Optimal Taxation and Public Production I: Production Efficiency and II: Tax Rules*, American Economic Review 61, 8–27 and 261–278.

A customs duty on HS 7601 is that instrument. Primary aluminium is not consumed; it is an input to extrusion, rolling, casting and fabrication. Taxing it alters the relative price of a produced input against labour, against capital, and against the finished imported article, which is untaxed.

3.2 Target the distortion at its source

Bhagwati and Ramaswami established that where the objective concerns domestic production, the first-best instrument acts directly on production; a trade instrument is second-best, because it attains the production objective only by simultaneously distorting consumption.² If the objective is to sustain domestic smelting capacity, the theorem prescribes a production subsidy, not an import duty.

3.3 The decisive institutional difference: creditability

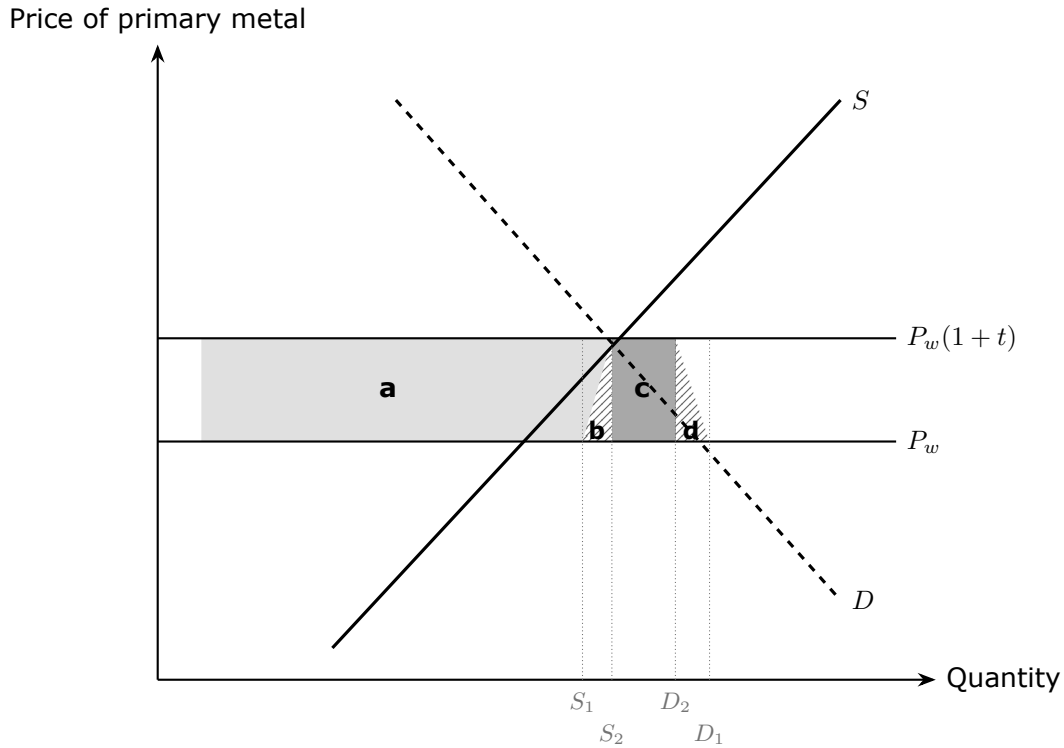
One fact settles the comparison in practice. **IGST paid on imported metal is creditable; basic customs duty is not.** A registered fabricator recovers the 18 per cent IGST against output liability, so it never enters cost. The 8.25 per cent duty is unrecoverable: it enters the cost base, is marked up at each subsequent stage, and cascades. Two nominally similar levies therefore behave in opposite ways.

4 Incidence: who gains and who pays

4.1 The diagram, with one adaptation

Figure 1 is the small-country tariff diagram applied to primary aluminium. The adaptation that matters is on the demand side: the buyers of HS 7601 are not households but *downstream producers*. What the textbook labels consumer surplus is producer surplus in the fabrication sector — the surplus whose employment and wage consequences Section 4 of the report measures.

²Bhagwati & Ramaswami (1963), *Domestic Distortions, Tariffs, and the Theory of Optimum Subsidy*, *Journal of Political Economy* 71(1), 44–50; see Panagariya (2006), *The World Economy* 29(10).



a — surplus gained by upstream producers. **c** — duty collected. **b** and **d** — deadweight loss. Downstream producers lose **a + b + c + d**; national welfare loses **b + d**. **The figure still flatters the duty**: it is drawn at an import share of about 20 per cent for legibility, whereas the actual share is 8.7 per cent, so area **c** is in reality narrower still relative to area **a**.

Figure 1: The incidence of an input duty under import-parity pricing

4.2 The transfer-to-revenue ratio is an identity

Areas *a* and *c* are each the price wedge multiplied by a quantity, so their ratio depends only on quantities:

$$a = \phi t P_w (Q_C - Q_M), \quad c = (1 - f) t P_w Q_M \quad \Rightarrow \quad \frac{a}{c} = \frac{\phi (1 - \mu)}{\mu (1 - f)}.$$

Price, exchange rate and both elasticities cancel. What remains is import penetration, pass-through and the duty-free share — and only the last of these is seriously uncertain.

Import penetration μ	$f = 0\%$	$f = 20\%$	$f = 40\%$	$f = 60\%$
5%	19.0	23.7	31.7	47.5
8.7% (baseline)	10.5	13.1	17.5	26.2
10%	9.0	11.2	15.0	22.5
15%	5.7	7.1	9.4	14.2
20%	4.0	5.0	6.7	10.0

Rupees transferred from downstream to upstream per rupee collected, at $\phi = 1$. At $\phi = 0.7$ every cell falls by 30 per cent; the most conservative cell in the table then becomes 2.8.

4.3 Levels, and the cost of raising a rupee this way

		INR crore
a	Upstream producer surplus gained	7,747
c	Duty collected, net of duty-free imports at the statutory rate on every tonne	444 739
b	Production deadweight	64
d	Conversion forgone	175
a+b+c+d	Downstream surplus lost	8,430
b+d	National welfare lost	239

Two ratios follow, and they are the numbers worth quoting.

For every rupee the exchequer collects through this duty:

- **INR 17.5** is transferred from downstream margins to upstream margins; and
- **INR 0.54** of value is destroyed outright, before counting the transfer.

A tax that destroys fifty-four paise of national income per rupee raised is an expensive way to raise revenue on any view. The corresponding figure for a creditable consumption tax on final goods is, by construction, far lower — which is the Diamond–Mirrlees point restated in rupees.

5 Replacing the revenue

5.1 What has to be replaced

	Net of duty-free imports	At the statutory maximum
Revenue to replace	INR 444 crore	INR 739 crore
GST rate change on the downstream base	+0.18 pp	+0.30 pp
i.e. from 18 per cent to	18.18%	18.30%
Share of FY26 gross GST collections	0.020%	0.033%
Days of GST collection	0.07	0.12

The base used is the ASI output of the three fabrication classes, INR 2.46 lakh crore. Applied to a wider aluminium-articles base the required change would be smaller still.

5.2 The induced-expansion offset

Removing an 8.25 per cent duty on metal raises downstream value added by $4.91 \times \theta \times t \times \phi$, where 4.91 is the amplification factor established in the report:

θ	Value added released	GST at 18% on it	vs INR 444 cr forgone
30%	+12.2% = INR 5,062 cr	INR 911 cr	covers it twice over
45%	+18.2% = INR 7,593 cr	INR 1,367 cr	covers it
60%	+24.3% = INR 10,124 cr	INR 1,822 cr	covers it
70%	+28.4% = INR 11,811 cr	INR 2,126 cr	covers it

Quote this table carefully. It is illustrative, not a forecast. It assumes the relief is retained in the sector rather than competed away, that capacity exists to convert it into output, and that no offsetting upstream contraction reduces the GST base elsewhere. What it establishes is not that the swap certainly pays for itself, but that plausible second-round effects are an order of magnitude larger than the first-round loss. A fiscal objection to a measure of this size is not a serious objection.

6 Sensitivity

The headline ratio was recomputed at every corner of the declared ranges for Q_C , ϕ , f and the exchange rate: a/c **runs from 5.5 to 39.2**, against a baseline of 17.5. It does not approach unity anywhere in the parameter space, which is the only property the argument depends on.

The absolute levels are less robust, as they should be. The transfer scales with ϕ and with Q_C ; the deadweight scales with the elasticities, which are drawn from the international metals literature rather than estimated for India. The deadweight figures should therefore be read as an order of magnitude — but note that they enter the argument only to make it *stronger*, and omitting them entirely still leaves the transfer-to-revenue ratio intact.

7 Does the evidence support revenue recovery?

Baunsgaard and Keen, on panel data for 117 countries over 32 years, find that high-income countries recover fully from domestic taxes what they lose from trade liberalisation; that middle-income countries show robust replacement, essentially dollar-for-dollar in the long run; and that recovery is weakest among low-income countries. They also find that the presence of a value-added tax has in itself made it easier to cope with the revenue effects of liberalisation.³ India is a middle-income economy with a mature, comprehensively administered VAT: it is squarely in the category for which the evidence of full replacement is strongest.

8 The counterarguments, and how far they go

8.1 Informality: the Emran–Stiglitz objection

The most serious theoretical challenge to substituting VAT for trade taxes is Emran and Stiglitz's.⁴ The standard prescription is derived from models with no informal sector; once the VAT's incomplete coverage is acknowledged, a revenue-neutral reduction in trade taxes offset by higher VAT can reduce welfare, because the trade tax reaches activity the VAT cannot.

The objection has force and should be stated rather than avoided. Two things limit its application here. First, it concerns taxes on *final consumption* where the informal sector escapes the net; the instrument at issue is a tariff on an *intermediate input* bought overwhelmingly by registered factories and cleared through customs.

³Baunsgaard & Keen (2010), *Tax revenue and (or?) trade liberalization*, Journal of Public Economics 94(9–10), 563–577.

⁴Emran & Stiglitz (2005), *On selective indirect tax reform in developing countries*, Journal of Public Economics 89(4), 599–623.

Diamond–Mirrlees applies to that case directly, and Emran–Stiglitz does not displace it. Second, a border levy functions in part as a withholding device on downstream informality — Keen’s argument — but IGST at the border already performs that function, and performs it without entering anybody’s cost base.⁵

8.2 The upstream adjustment

Rationalisation is not costless upstream, and the note should not pretend otherwise. Domestic realisations would fall towards import parity, and area a — about INR 7,750 crore on the baseline — is the measure of what is lost. Two considerations bound it: the segment is currently earning substantial operating surpluses and expanding capacity, so the change compresses margin rather than threatening viability; and a large share of Indian primary output is exported at world prices already, and is unaffected.

8.3 The trade-agreement constraint

A reduction in the applied MFN rate extends to all partners, so the measure cannot be targeted. That bears on *how* to rationalise rather than *whether*: it points towards a general reduction rather than a partner-specific one, and it is why the report treats trade-remedy instruments as a separate question from the input duty.

9 How to frame this in the report

1. **Lead with the identity, not the levels.** $a/c = \phi(1 - \mu)/[\mu(1 - f)]$ requires no price, no exchange rate and no elasticity. A reader can check it on the back of an envelope, and there is nothing in it to dispute except import penetration, which is published.
2. **Give the two per-rupee numbers together.** “INR 17.5 transferred and 54 paise destroyed, per rupee collected” is the sentence that carries the section.
3. **Put the instrument comparison before the arithmetic.** A reader who has accepted that BCD sticks to cost while GST does not will accept what follows.
4. **Name the theorems.** Diamond–Mirrlees and Bhagwati–Ramaswami convert what sounds like an industry request into the textbook treatment of an input tax.
5. **Concede the upstream loss and quantify it.** Area a is the concession. Making it with a number attached is what distinguishes this from a lobbying document.
6. **State the Emran–Stiglitz objection first, then answer it.**
7. **Do not overclaim §5.2.** The defensible claim is that the sum at stake is too small to be decisive — not that reform is costless.

10 Suggested placement

A new subsection in Section 5, immediately before the policy recommendations, running to about a page and a half: the instrument comparison, the identity and its table, the two per-rupee ratios, and the GST equivalence. The baseline table, the welfare

⁵Keen (2008), *VAT, tariffs, and withholding: Border taxes and informality in developing countries*, *Journal of Public Economics* 92(10–11), 1892–1906.

accounting, the sensitivity analysis and the counterargument discussion belong in an appendix, consistent with the convention already adopted for the ASI material.

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